SNORT IDS – Raspberry Pi

Group 1 – Cyberchase

CYEN 3331 – Computer Network Security

Houston Christian University

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Initial Setup of Raspberry Pi and Kali Linux

The first step in this lab was preparing the IDS/IPS system using a Raspberry Pi. Kali Linux was installed on a microSD card using the Raspberry Pi imaging tool, following the provided instructions. After installation, the SD card was inserted into the Raspberry Pi and the device was powered on. A headless configuration was not used. A monitor and keyboard was used to enable SSH.

Network Configuration and Remote Access

After booting the Raspberry Pi, we identified its IP address on the network and connected to it using SSH. Once connected, we configured the Wi-Fi setting and verified that the device was properly connected to the network. This step ensured that the Raspberry Pi could act as the IDS/IPS monitoring system within the network.

Installation of Snort

Snort was installed using the package manager. The system repositories were updated, and Snort along with its dependencies was successfully installed. We verified the installation by running Snort commands and confirming that the configuration file (snort.lua) loaded correctly without errors. This confirmed that Snort was properly installed and ready for configuration.

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo apt upgrade -y
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo apt install snort -y
The following packages were automatically installed and are no longer require
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ snort -V

   ,,_     -*> Snort++ <*-
  o"  )~   Version 3.12.1.0
   ''''    By Martin Roesch & The Snort Team
           http://snort.org/contact#team
           Copyright (C) 2014-2026 Cisco and/or its affiliates. All rights re
served.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using DAQ version 3.0.24
           Using libpcap version 1.10.6 (64-bit time_t, with TPACKET_V3)
           Using LuaJIT version 2.1.1761786044
           Using LZMA version 5.8.2
           Using OpenSSL 3.5.5 27 Jan 2026
           Using PCRE2 version 10.46 2025-08-27
           Using ZLIB version 1.3.1
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo snort -c /etc/snort/snort.lua
--------------------------------------------------
o")~   Snort++ 3.12.1.0
--------------------------------------------------
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
        ips
        alerts
        daq
        decode
        host_cache
        host_tracker
        hosts
        network
        packets
        process
        search_engine
        stream
        stream_ip
        stream_icmp
        stream_tcp
        stream_udp
        stream_user
        stream_file
        arp_spoof
        back_orifice
        dns
        imap
        rpc_decode
        sip
        socks
        ssh
        ssl
        telnet
        cip
        dnp3
        iec104
        mms
```

```
        mms
        modbus
        opcua
        s7commplus
        dce_smb
        dce_tcp
        dce_udp
        dce_http_server
        gtp_inspect
        port_scan
        smtp
        ftp_server
        ftp_client
        ftp_data
        http_inspect
        output
        file_policy
        appid
        wizard
        binder
        classifications
        js_norm
        file_inspect
        http2_inspect
        dce_http_proxy
        pop
        normalizer
        netflow
        references
        active
Finished /etc/snort/snort.lua:
Loading file_inspect.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_inspect.rules_file:
--------------------------------------------------
ips policies rule stats
               id  loaded  shared enabled    file
                1     219       0     219    /etc/snort/snort.lua
--------------------------------------------------
```

```
-------------------------------------------------
rule counts
       total rules loaded: 219
               text rules: 219
            option chains: 219
            chain headers: 1
-------------------------------------------------
service rule counts          to-srv  to-cli
                  file_id:      219     219
                    total:      219     219
-------------------------------------------------
fast pattern groups
                to_server: 1
                to_client: 1
-------------------------------------------------
search engine (ac_bnfa)
                instances: 2
                 patterns: 438
            pattern chars: 2602
               num states: 1832
         num match states: 392
             memory scale: KB
             total memory: 71.2812
           pattern memory: 19.6484
        match list memory: 28.4375
        transition memory: 22.9453
appid: MaxRss diff: 2816
appid: patterns loaded: 300
-------------------------------------------------
pcap DAQ configured to passive.

Snort successfully validated the configuration (with 0 warnings).
o")~   Snort exiting
```

Configuration of Snort and Rules

Next, we created a custom rule to detect traffic. The rule was added to the local.rules file in the Snort rules directory. This rule was designed to generate alerts whenever ICMP echo requests were detected. We validated the configuration using Snort's testing command, which confirmed that there were no errors or warning in the configuration.

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo nano /etc/snort/rules/local.rules
```

```
  GNU nano 8.7.1                    /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# --------------
# LOCAL RULES
# --------------
# This file intentionally does not come with signatures.  Put your local
# additions here.

alert icmp any any -> any any (msg:"ICMP Echo Request detected"; sid:1000001;)
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules
--------------------------------------------------
o")~   Snort++ 3.12.1.0
--------------------------------------------------
```

```
--------------------------------------------------
search engine (ac_bnfa)
                instances: 2
                 patterns: 438
            pattern chars: 2602
               num states: 1832
         num match states: 392
             memory scale: KB
             total memory: 71.2812
           pattern memory: 19.6484
        match list memory: 28.4375
        transition memory: 22.9453
appid: MaxRss diff: 2944
appid: patterns loaded: 300
--------------------------------------------------
pcap DAQ configured to passive.

Snort successfully validated the configuration (with 0 warnings).
o")~   Snort exiting
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules -i wlan0 -A alert_fast
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ ping 10.0.0.225
PING 10.0.0.225 (10.0.0.225) 56(84) bytes of data.
64 bytes from 10.0.0.225: icmp_seq=1 ttl=64 time=0.098 ms
64 bytes from 10.0.0.225: icmp_seq=2 ttl=64 time=0.089 ms
64 bytes from 10.0.0.225: icmp_seq=3 ttl=64 time=0.099 ms
64 bytes from 10.0.0.225: icmp_seq=4 ttl=64 time=0.097 ms
^C
--- 10.0.0.225 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.089/0.095/0.099/0.004 ms
```

```
------------------------------------------------------------
fast pattern groups
                to_server: 1
                to_client: 1
------------------------------------------------------------
search engine (ac_bnfa)
                instances: 2
                 patterns: 438
            pattern chars: 2602
               num states: 1832
          num match states: 392
             memory scale: KB
             total memory: 71.2812
           pattern memory: 19.6484
         match list memory: 28.4375
         transition memory: 22.9453
appid: MaxRss diff: 3072
appid: patterns loaded: 300
------------------------------------------------------------
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] wlan0
04/04-18:11:46.543985 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::a
edb:48ff:fe6f:bd41 -> ff02::1
04/04-18:11:46.557015 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::8
289:f95f:d40e:f4ba -> ff02::16
04/04-18:11:47.248979 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::8
289:f95f:d40e:f4ba -> ff02::16
```

```
04/04-18:12:14.292984 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::8
289:f95f:d40e:f4ba -> ff02::16
04/04-18:12:16.549963 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::aedb:48ff:fe6f:
bd41 -> ff02::1
04/04-18:12:16.560998 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::8289:f95f:d40e:
f4ba -> ff02::16
04/04-18:12:16.976988 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::8289:f95f:d40e:
f4ba -> ff02::16
04/04-18:12:19.624118 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::aedb:48ff:fe6f:
bd41 -> ff02::1
04/04-18:12:19.636997 [**] [1:1000001:0] "ICMP Echo Request detected" [**] [Priority: 0] {ICMP} fe80::8289:f95f:d40e:
f4ba -> ff02::16
```

Running Snort in Monitoring Mode

Snort was the executed in live monitoring mode using the appropriate flags to specify the configuration file, load the custom rules, monitor the network interface, and display alerts. To test the IDS functionality, we attempted to generate traffic and generate alerts based on the rule we created.

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ wget https://www.snort.org/downloads/community/snort3-community-rules.tar.gz
--2026-04-04 18:17:58--  https://www.snort.org/downloads/community/snort3-community-rules.tar.gz
Resolving www.snort.org (www.snort.org)... 2606:4700::6810:5c13, 2606:4700::6810:5b13, 104.16.91.19,
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ tar -xvzf snort3-community-rules.tar.gz
snort3-community-rules/
```

Move the rules to Snort's rules directory:

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo mv snort3-community-rules /etc/snort/rules
```

Running Snort with Community Rules:

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/snort3-community-rules -i wlan0 -A alert_fast
--------------------------------------------------
o")~   Snort++ 3.12.1.0
--------------------------------------------------
```

```
kali@kali-rasp
Session  Actions  Edit  View  Help
service rule counts          to-srv  to-cli
                  file_id:      219     219
                    total:      219     219
fast pattern groups
                to_server: 1
                to_client: 1
search engine (ac_bnfa)
                instances: 2
                 patterns: 438
            pattern chars: 2602
               num states: 1832
         num match states: 392
             memory scale: KB
             total memory: 71.2812
           pattern memory: 19.6484
        match list memory: 28.4375
        transition memory: 22.9453
appid: MaxRss diff: 3240
appid: patterns loaded: 300
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] wlan0
```

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/snort3-community-rules -i wlan0 -A alert_
fast -q -l /var/log/snort
```

Verify Snort is working on our network

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 2c:cf:67:4d:dc:29 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::7cf5:111f:dc6e:3308/64 scope link
       valid_lft forever preferred_lft forever
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 2c:cf:67:4d:dc:2a brd ff:ff:ff:ff:ff:ff
    inet 172.20.10.2/28 brd 172.20.10.15 scope global dynamic noprefixroute wlan0
       valid_lft 2594sec preferred_lft 2594sec
    inet6 2607:fb90:5d62:b8c:174c:ebde:d35e:20dc/64 scope global noprefixroute
       valid_lft forever preferred_lft forever
    inet6 fe80::43a1:50a:18b:767e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo ip addr flush dev eth0

┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo ip addr add 10.10.1.11/24 dev eth0

┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo ip route add default via 10.10.1.1
```

After rebooting verifying the ip

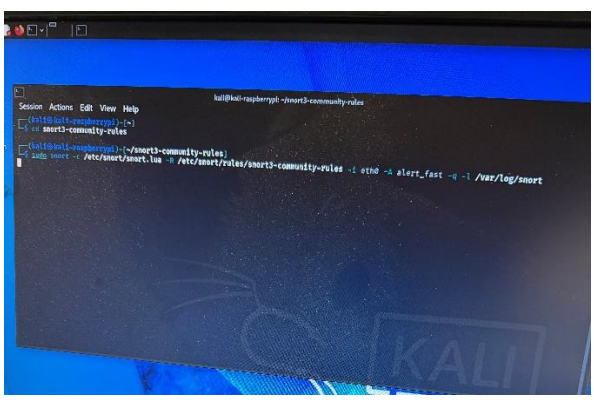
```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 2c:cf:67:4d:dc:29 brd ff:ff:ff:ff:ff:ff
    inet 169.254.133.111/16 brd 169.254.255.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
    inet 10.10.1.11/24 scope global eth0
       valid_lft forever preferred_lft forever
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 2c:cf:67:4d:dc:2a brd ff:ff:ff:ff:ff:ff
    inet 172.20.10.2/28 brd 172.20.10.15 scope global dynamic noprefixroute wlan0
       valid_lft 3288sec preferred_lft 3288sec
    inet6 2607:fb90:5d62:b8c:174c:ebde:d35e:20dc/64 scope global noprefixroute
       valid_lft forever preferred_lft forever
    inet6 fe80::43a1:50a:18b:767e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

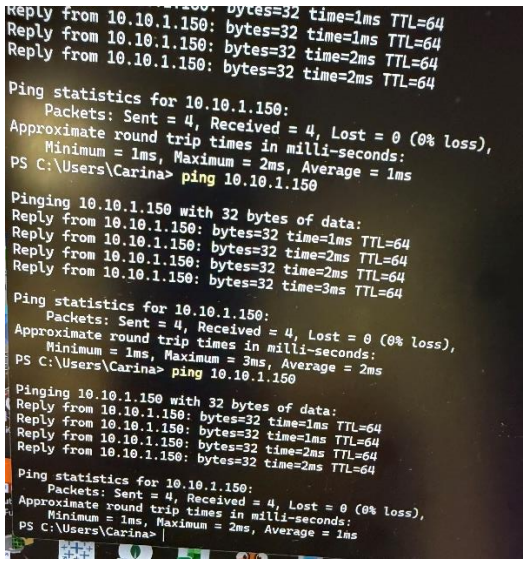
Test if switch mirroring works

```
┌──(kali㉿kali-raspberrypi)-[~]
└─$ sudo tcpdump -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:54:43.809327 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:44.124565 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 80:3f:5d:81:1e:
17 (oui Unknown), length 300
14:54:44.833171 Loopback, skipCount 0, Reply, receipt number 0, data (40 octets)
14:54:45.822103 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:46.290871 IP 10.10.1.1 > 224.0.0.5: OSPFv2, Hello, length 56
14:54:47.305769 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 80:3f:5d:81:1e:
17 (oui Unknown), length 300
14:54:47.831550 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:49.834805 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:50.338915 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 80:3f:5d:81:1e:
17 (oui Unknown), length 300
14:54:51.837501 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:53.380044 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 80:3f:5d:81:1e:17 (oui Un
known), length 300
14:54:53.840943 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:54.838172 Loopback, skipCount 0, Reply, receipt number 0, data (40 octets)
14:54:55.843748 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:56.024183 IP 10.10.1.1 > 224.0.0.5: OSPFv2, Hello, length 56
14:54:56.037046 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 80:3f:5d:81:1e:17 (oui Un
known), length 300
14:54:57.846699 STP 802.1d, Config, Flags [none], bridge-id 8001.d4:d7:48:3e:83:00.801f, length 35
14:54:59.078192 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 80:3f:5d:81:1e:17 (oui Un
```

Issue with Traffic Detection and Logging

Steps 3a-e were skipped. Despite attempting to generate traffic, the Pi was unable to observe any alerts or logged activity in Snort. No traffic appeared to be captured by the IDS system. It was determined that the issue was likely due to the switch port mirroring configuration not being completed or some other issue with the Pi. A TA was unable to observe any alerts or logged activity despite generating traffic.





In summary, all setup and configuration steps for Snort were completed successfully, including installation, rule creation, and execution. However, due to the

missing network traffic mirroring configuration, the IDS system was unable to capture or log traffic. As a result, the final objective of detecting and logging network activity could not be achieved.

Span_switch_setup

```
S1>en

Password:

S1#show vlan brief


VLAN Name                             Status    Ports

---- -------------------------------- --------- -------------------------------

1    default                          active    Gi1/0/1, Gi1/0/2, Gi1/0/3

                                                Gi1/0/4, Gi1/0/5, Gi1/0/6

                                                Gi1/0/7, Gi1/0/8, Gi1/0/9

                                                Gi1/0/10, Gi1/0/11, Gi1/0/12

                                                Gi1/0/13, Gi1/0/14, Gi1/0/15

                                                Gi1/0/16, Gi1/0/17, Gi1/0/18

                                                Gi1/0/19, Gi1/0/20, Gi1/0/21

                                                Gi1/0/22, Gi1/0/23, Gi1/0/24

                                                Gi1/0/25, Gi1/0/26, Gi1/0/27

                                                Gi1/0/28, Gi1/0/29, Gi1/0/30

                                                Gi1/0/31, Gi1/0/32, Gi1/0/33

                                                Gi1/0/34, Gi1/0/35, Gi1/0/36

                                                Gi1/0/37, Gi1/0/38, Gi1/0/39

                                                Gi1/0/40, Gi1/0/41, Gi1/0/42

                                                Gi1/0/43, Gi1/0/44, Gi1/0/45

                                                Gi1/0/46, Gi1/0/47, Gi1/0/48

                                                Gi1/0/49, Gi1/0/50, Gi1/0/51

                                                Gi1/0/52
```

```
999  VLAN0999                         active
1002 fddi-default                     act/unsup
 --More--
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
S1#  conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#int Gi1/0/2
S1(config-if)#switchportmode      mode cce   access
S1(config-if)#no shut
S1(config-if)#end
S1#sh
*Jan  2 00:04:01.754: %SYS-5-CONFIG_I: Configured from console by consolew
S1#show running-config
Building configuration...

Current configuration : 4213 bytes
!
! Last configuration change at 00:04:01 UTC Mon Jan 2 2006
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
```

```
service password-encryption
!
hostname S1
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$JLMs$pVXW.6hkUu3q6boeSBvDf/
!
no aaa new-model
switch 1 provision ws-c2960s-48fps-l
!
!
!
 --More--         !
crypto pki trustpoint TP-self-signed-1212056320
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1212056320
 revocation-check none
 rsakeypair TP-self-signed-1212056320
!
!
crypto pki certificate chain TP-self-signed-1212056320
 certificate self-signed 01
  3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
  31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
```

69666963 6174652D 31323132 30353633 3230301E 170D3036 30313032 30303032

31395A17 0D323030 31303130 30303030 305A3031 312F302D 06035504 03132649

4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 32313230

35363332 3030819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281

8100A744 5A46C621 D9D47E3D DA3160B5 9896C5A0 A34033FA 0810D0A1 C12DA5EC

4C0E3869 8DF3F7AA 854C3C53 0A7EE0EE 3FA3594C 5E3F3504 BBF26692 8E5614CD

B108B4D7 707F1623 484660B2 A22F2A8B 1BCBB808 C80AF363 9D0F5F0B E016BBE4

CA405BC4 FEF42D06 2E331DA4 13448722 E13A4345 769D527D A363275D C5013F5A

70990203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603

551D2304 18301680 14918849 55DA7665 4634C5EF F92B66D7 419FB652 01301D06

03551D0E 04160414 91884955 DA766546 34C5EFF9 2B66D741 9FB65201 300D0609

--More-- 2A864886 F70D0101 05050003 8181008D 0DF7B749 8849CCC8 D98A2DC2 5F2E8702

0550E0BC 725785A0 A9A98001 0E0A8533 00276266 B54F675F 8BDAF4B9 A23DAFDE

28F8EE5C 84C611F6 50C9F123 50D0A5CD F02EBC88 48BBFF49 F8451EA8 EB31113B

189A553D 19C17B73 FB547E37 59C6B17B 8083658D CBD47BC2 152FCC39 061DAE21

D9C4B340 A1D6ECFC 4B6AA3B6 4D0AC8

quit

spanning-tree mode pvst

spanning-tree extend system-id

!

```
!
!
!
!
!
!
!
!
vlan internal allocation policy ascending
!
!
!
!
!
 --More--         !
!
!
!
!
!
interface FastEthernet0
 no ip address
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
 switchport mode access
```

```
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
 --More--         !
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
```

```
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
 --More--        interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/0/25
!
interface GigabitEthernet1/0/26
!
interface GigabitEthernet1/0/27
!
interface GigabitEthernet1/0/28
!
interface GigabitEthernet1/0/29
```

```
!
interface GigabitEthernet1/0/30
 --More--        !
interface GigabitEthernet1/0/31
!
interface GigabitEthernet1/0/32
!
interface GigabitEthernet1/0/33
!
interface GigabitEthernet1/0/34
!
interface GigabitEthernet1/0/35
!
interface GigabitEthernet1/0/36
!
interface GigabitEthernet1/0/37
!
interface GigabitEthernet1/0/38
!
interface GigabitEthernet1/0/39
!
interface GigabitEthernet1/0/40
!
interface GigabitEthernet1/0/41
!
 --More--        interface GigabitEthernet1/0/42
!
```

```
interface GigabitEthernet1/0/43
!
interface GigabitEthernet1/0/44
!
interface GigabitEthernet1/0/45
!
interface GigabitEthernet1/0/46
!
interface GigabitEthernet1/0/47
!
interface GigabitEthernet1/0/48
!
interface GigabitEthernet1/0/49
!
interface GigabitEthernet1/0/50
!
interface GigabitEthernet1/0/51
!
interface GigabitEthernet1/0/52
!
interface Vlan1
 --More--        ip address 10.10.1.100 255.255.255.0
!
ip http server
ip http secure-server
!
!
```

```
!
vstack
banner motd ^CWarning Unauthorized Access Prohibited!^C
!
line con 0
 password 7 05080F1C2243
 login
line vty 0 4
 login
line vty 5 15
 login
!
!
monitor session 1 source interface Gi1/0/31
end

S1# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#no monitor session 1
S1(config)#no  monitor sesis  ssion 1        ssion 1         ssion 1 sp ource vlan 1 both
S1(config)#monitor session 1 destination interface Gi1/0/2
S1(config)#noi shutdown
% Incomplete command.

S1(config)#no shut
% Incomplete command.
```

```
S1(config)#end
S1#
*Jan  2 00:05:29.090: %SYS-5-CONFIG_I: Configured from console by consoles
S1#show monitor
Session 1
---------
Type                    : Local Session
Source VLANs            :
    Both                : 1
Destination Ports       : Gi1/0/2
    Encapsulation       : Native
          Ingress       : Disabled


S1#show mi onitor session 1
Session 1
---------
Type                    : Local Session
Source VLANs            :
    Both                : 1
Destination Ports       : Gi1/0/2
    Encapsulation       : Native
          Ingress       : Disabled


S1#exit
```

S1 con0 is now available

Press RETURN to get started.

Span_startup_config

=~=~=~=~=~=~=~=~=~=~=~=~= PuTTY log 2026.04.08 13:02:06
=~=~=~=~=~=~=~=~=~=~=~=~=

Warning Unauthorized Access Prohibited!

User Access Verification

Password:

S1>en

```
Password:
S1#show running  -config
Building configuration...

Current configuration : 4249 bytes
!
! Last configuration change at 00:05:29 UTC Mon Jan 2 2006
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname S1
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$JLMs$pVXW.6hkUu3q6boeSBvDf/
!
no aaa new-model
switch 1 provision ws-c2960s-48fps-l
!
!
!
 --More--         !
```

```
crypto pki trustpoint TP-self-signed-1212056320
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1212056320
 revocation-check none
 rsakeypair TP-self-signed-1212056320
!
!
crypto pki certificate chain TP-self-signed-1212056320
 certificate self-signed 01
  3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
  31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 31323132 30353633 3230301E 170D3036 30313032 30303032
  31395A17 0D323030 31303130 30303030 305A3031 312F302D 06035504 03132649
  4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 32313230
  35363332 3030819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
  8100A744 5A46C621 D9D47E3D DA3160B5 9896C5A0 A34033FA 0810D0A1 C12DA5EC
  4C0E3869 8DF3F7AA 854C3C53 0A7EE0EE 3FA3594C 5E3F3504 BBF26692 8E5614CD
  B108B4D7 707F1623 484660B2 A22F2A8B 1BCBB808 C80AF363 9D0F5F0B E016BBE4
  CA405BC4 FEF42D06 2E331DA4 13448722 E13A4345 769D527D A363275D C5013F5A
  70990203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
  551D2304 18301680 14918849 55DA7665 4634C5EF F92B66D7 419FB652 01301D06
  03551D0E 04160414 91884955 DA766546 34C5EFF9 2B66D741 9FB65201 300D0609
```

--More-- 2A864886 F70D0101 05050003 8181008D 0DF7B749 8849CCC8 D98A2DC2 5F2E8702

0550E0BC 725785A0 A9A98001 0E0A8533 00276266 B54F675F 8BDAF4B9 A23DAFDE

28F8EE5C 84C611F6 50C9F123 50D0A5CD F02EBC88 48BBFF49 F8451EA8 EB31113B

189A553D 19C17B73 FB547E37 59C6B17B 8083658D CBD47BC2 152FCC39 061DAE21

D9C4B340 A1D6ECFC 4B6AA3B6 4D0AC8

quit

spanning-tree mode pvst

spanning-tree extend system-id

!

!

!

!

!

!

!

!

!

vlan internal allocation policy ascending

!

!

!

!

!

--More-- !

```
!
!
!
!
!
interface FastEthernet0
 no ip address
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
 switchport mode access
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
 --More--         !
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
```

```
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
 --More--        interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
```

```
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/0/25
!
interface GigabitEthernet1/0/26
!
interface GigabitEthernet1/0/27
!
interface GigabitEthernet1/0/28
!
interface GigabitEthernet1/0/29
!
interface GigabitEthernet1/0/30
 --More--         !
interface GigabitEthernet1/0/31
!
interface GigabitEthernet1/0/32
!
interface GigabitEthernet1/0/33
!
interface GigabitEthernet1/0/34
!
interface GigabitEthernet1/0/35
!
interface GigabitEthernet1/0/36
!
```

```
interface GigabitEthernet1/0/37
!
interface GigabitEthernet1/0/38
!
interface GigabitEthernet1/0/39
!
interface GigabitEthernet1/0/40
!
interface GigabitEthernet1/0/41
!
 --More--        interface GigabitEthernet1/0/42
!
interface GigabitEthernet1/0/43
!
interface GigabitEthernet1/0/44
!
interface GigabitEthernet1/0/45
!
interface GigabitEthernet1/0/46
!
interface GigabitEthernet1/0/47
!
interface GigabitEthernet1/0/48
!
interface GigabitEthernet1/0/49
!
interface GigabitEthernet1/0/50
```

```
!
interface GigabitEthernet1/0/51
!
interface GigabitEthernet1/0/52
!
interface Vlan1
 --More--        ip address 10.10.1.100 255.255.255.0
!
ip http server
ip http secure-server
!
!
!
vstack
banner motd ^CWarning Unauthorized Access Prohibited!^C
!
line con 0
 password 7 05080F1C2243
 login
line vty 0 4
 login
line vty 5 15
 login
!
!
monitor session 1 source vlan 1
monitor session 1 destination interface Gi1/0/2
```

```
end

S1#copy running  -config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#copy running-config startup-config                    show startup-config
Using 2955 out of 524288 bytes
!
! Last configuration change at 00:05:29 UTC Mon Jan 2 2006
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname S1
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$JLMs$pVXW.6hkUu3q6boeSBvDf/
!
no aaa new-model
switch 1 provision ws-c2960s-48fps-l
!
```

```
!
!
!
 --More--         crypto pki trustpoint TP-self-signed-1212056320
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1212056320
 revocation-check none
 rsakeypair TP-self-signed-1212056320
!
!
crypto pki certificate chain TP-self-signed-1212056320
 certificate self-signed 01 nvram:IOS-Self-Sig#2.cer
spanning-tree mode pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
vlan internal allocation policy ascending
!
!
 --More--         !
```

```
!
!
!
!
!
!
!
!
interface FastEthernet0
 no ip address
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
 switchport mode access
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
 --More--        interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
```

```
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
 --More--         !
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
```

```
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/0/25
!
interface GigabitEthernet1/0/26
!
interface GigabitEthernet1/0/27
!
interface GigabitEthernet1/0/28
!
 --More--        interface GigabitEthernet1/0/29
!
interface GigabitEthernet1/0/30
!
interface GigabitEthernet1/0/31
!
interface GigabitEthernet1/0/32
!
interface GigabitEthernet1/0/33
!
interface GigabitEthernet1/0/34
!
interface GigabitEthernet1/0/35
```

```
!
interface GigabitEthernet1/0/36
!
interface GigabitEthernet1/0/37
!
interface GigabitEthernet1/0/38
!
interface GigabitEthernet1/0/39
!
interface GigabitEthernet1/0/40
 --More--         !
interface GigabitEthernet1/0/41
!
interface GigabitEthernet1/0/42
!
interface GigabitEthernet1/0/43
!
interface GigabitEthernet1/0/44
!
interface GigabitEthernet1/0/45
!
interface GigabitEthernet1/0/46
!
interface GigabitEthernet1/0/47
!
interface GigabitEthernet1/0/48
!
```

```
interface GigabitEthernet1/0/49
!
interface GigabitEthernet1/0/50
!
interface GigabitEthernet1/0/51
!
 --More--         interface GigabitEthernet1/0/52
!
interface Vlan1
 ip address 10.10.1.100 255.255.255.0
!
ip http server
ip http secure-server
!
!
!
vstack
banner motd ^CWarning Unauthorized Access Prohibited!^C
!
line con 0
 password 7 05080F1C2243
 login
line vty 0 4
 login
line vty 5 15
 login
!
```

```
!
monitor session 1 source vlan 1
 --More--         monitor session 1 destination interface Gi1/0/2
end

S1#
S1#exit
```

S1 con0 is now available

Press RETURN to get started.